Investigating discounting and cooperation in a simple threshold public goods game

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Background

Public goods, that is goods that offer benefits that are nonexcludable and nonrival, are both common and important in our societies. Some notable examples include climate change mitigation, disease eradication, herd immunity from vaccination programs, and programs such as national defense. Hence, the study of public goods games is motivated by their practical application to societal problems. When provided successfully, public goods can make every party better off because no other population can be prevented from enjoying the public good and thus, optimal allocation and division of public goods is desirable since it makes everyone better off in a fair division of resources (Barrett, 2007). The major research interest in public goods games comes from investigating the conflict between individual and collective interest in group situations. Since there is a cost to cooperation, an individual who defects and does not contribute can free ride off of the contributions of others for a better individual payoff. This can lead to a dilemma where even though the group as a whole would be better off if all players cooperated, the payoff for a defector is higher than for a cooperator, so it is in an individual's best interest to defect (Wang, Fu, Wu , & Wang, 2009).

One particularly relevant type of public goods game is the threshold public goods game. In this situation, players must reach a collective target or threshold in order for the public good to be provided, which is analogous to real-life social situations where, for example, a certain amount of funding needs to be raised to launch a new public project (Bolle, 2014). Threshold public goods games are particularly interesting to study for a few reasons. First, they often have multiple equilibria, including cooperative equilibria, and experimental results have found that people are sometimes cooperative under these sorts of conditions, which means they can be useful for understanding human cooperative behavior (Wang et al., 2009). Second, in situations with threshold public goods in the real world there is generally a time delay between individuals' contributions towards the threshold and the launch of the public good, which provides an opportunity to investigate the effect of time discounting. This is particularly valuable because, as Jacquet et al. assert, more research is needed to understand the effects of temporal discounting on group behavior (Jacquet et al., 2013).

Discounting and other methods of reweighting future resources and benefits are tied to the idea that future utility obtains some lesser value at the present, discounted by a given weighting. The amount

we discount by depends upon our expectations about the quality of the environment, as well as our expectations of the certainty of the utility and the level of economic development in our future resources. This question is complex because of the myriad of parameters involved in the decision-making: from the considerable uncertainties faced with respect to both the ecological and economic evolutions of societies. In practical examples, utilities or payoffs are generally discounted by a factor that individuals expect a given utility to be able to grow at. In financial applications, discounting is used in loan payments or cash flows to reflect the riskiness associated with the payoffs, as well as investor expectations on the potential growth of the payments in a given period of time.

Previously, Hauert, Michor, Nowak and Doebeli (2006) investigated methods of discounting cooperation in social dilemmas. Using the concepts of discounting and synergy for describing how benefits accumulate when more than one cooperator is present in a group of interacting individuals, they demonstrated a few scenarios of evolutionary dynamics in which cooperators and defectors do not invade each other. In order to model the synergistically enhanced as well as the discounted net value of accumulated cooperative benefits as they relate to many societal resource problems, they assume that defectors and cooperators receive the following payoffs in a group with k cooperators:

$P_D(k) = b(1+ w + w^2+ \ldots + + w^{k-1})/N$, which is then summed as a geometric sequence to be $b(1-w^k)/(N(1-w))$.

So the first cooperator provides a benefit b which is shared by all N members of the group (including itself), w represents the weight or discount factor in a given time period, so w^j presents the discounting of period j to present value and the second one increases everyone's benefit by $b/N * w$ and so on, to the last of the k cooperators in the group providing an additional benefit of $b/N * w^{k-1}$. The cost of cooperation c, however, incur only to cooperators. If $w = 1$, then all cooperators provide the same incremental benefit b/N. This corresponds to the traditional formulation of Public Goods games with $P_D(k) = rkc/N$, where r denotes the multiplication factor of the common pool. If $w < 1$, then the benefits are discounted and the value of the benefits provided by each additional cooperator is lower than the previous. If $w > 1$, then the benefits are synergistically enhanced, and each additional cooperator provides incremental benefits of increasing value. It is important to note that neither discounting nor synergy refer to potential future benefits in iterated interactions (Fudenberg and Maskin, 1986).

The methods we consider in this paper are hyperbolic and exponential discounting. Hyperbolic discounting is a time-inconsistent model of delay discounting, that is the current relative valuation placed on receiving a good at an earlier date compared with receiving it at a later date. In hyperbolic discounting, valuations fall relatively rapidly for earlier delay periods, but then fall more slowly for longer delay periods. A hyperbolic discounting function can be described mathematically as $g(D) = 1/(1+kD)$, where

g(D) is the discount factor that multiplies the value of the reward, D is the delay in the reward, and k is a parameter governing the degree of discounting. On the contrary, exponential discounting is not dynamically inconsistent, meaning that the discount weighting factor is consistent over independent periods of time, and the discount weight on a payoff at a given time, will be as a function of time from the present.

Our paper investigates the role of time discounting in a specific threshold public goods game scenario: the game the class played in lecture during the first half of the semester. This is interesting because it provides a novel game as well as empirical data as a lens through which to address the broader problem of cooperation and discounting. In our specific investigation, we first work to understand the game we played in class through a simple pure-strategies game theoretical framework, then extend this framework to an extensive game model where we can investigate time discounting in two ways. By varying the discounting strength, we aim to understand how discounting affects the existence of cooperative behavior in this threshold public goods game. We also aim to compare exponential and hyperbolic discounting, and to use the empirical data from the class game as a framework to compare the model to the real world and to try to draw conclusions about the nature of discounting in real scenarios.

Introduction to the Class Experiment

During this semester, the class split into 4 groups to play a Threshold Public goods game. Two of the groups (B and C) played a modified version of the game, so we will only consider the simple form of the game (which was played by groups A and B). The basic structure of the class game was as follows: a team of 9 players were each given 2 chocolates per lecture for a total length of 10 lectures (5 weeks). At each lecture, players were given the option to contribute 0,1, or 2 chocolates to the group account, with a threshold goal of 90 chocolates, above which all players would be rewarded equally with 20 chocolates, or the equivalent of taking 2 chocolates for yourself every turn. At the end of the game, players filled out a short survey about their motivations and actions during the game. When we ran the game in class, group B did not reach the threshold (ended with 64 points) and group A did (with 98 points, Kruger, Barfuss, Vasconcelos, 2020). In this section of the project, we describe the game mathematically, which means understanding what are the possible strategies and outcomes of the game.

First, we look at the game as a normal form game where a player chooses a pure strategy of giving back x chocolates at the beginning of the game and always gives back x chocolates at each turn. In this framework, it is possible to build straightforward payoff matrices and look for potential Nash Equilibria and evolutionarily stable strategies directly. When we assume everyone in the game plays the same strategy at every turn, and all players choose the same strategy (except our focal player) out of

cooperation, ½ cooperation, and defection, we get the following payoff matrix for one individual (payoffs are to the row player).

	Everyone else contributes 2 chocolates	Everyone else contributes 1, keeps one	Everyone else withholds both chocolates
Contribute 2 chocolates	0+20	0+20	0+0
Contribute 1, keep one	10+20	10+20	10+0
Withhold both chocolates	20+20	20+0	20+0

Figure 1: *Simple payoff matrix for class game. The various strategies and corresponding payoffs for each player are shown. Based on (Kruger, Barfuss, Vasconcelos, 2020).*

Here we see that two equilibria are when everyone withholds both chocolates, or when everyone contributes one and withholds one (Figure 1). Otherwise, some player has an incentive to defect and hold onto their chocolates.

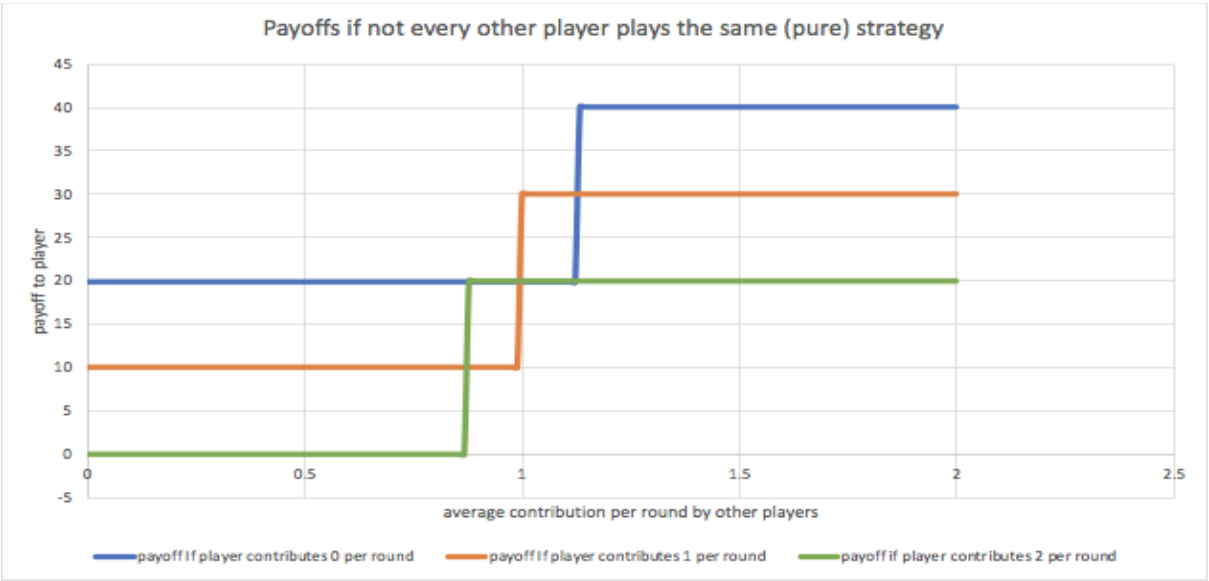

Figure 2: *Payoff to a player for each pure strategy vs other players' average contribution per round. We plot the average contribution per round of the rest of the group against the payoff from each strategy of the participant we are tracking.*

Since in the game we are playing with 9 players, not all of whom might value chocolate/value the group in the same way, it makes sense to investigate what happens when each player is allowed to pick a different pure strategy (still applying this strategy to each turn of the whole game). This produces the figure above (Figure 2), where we can see that contributing 0 is generally the best strategy, except for in a small window where the rest of the team contributes at or just above 1 chocolate on average, where it is best to contribute 1 chocolate to the group, and a small window just below 1 chocolate on average where the (non-discounted) payoff is the same for contributing both or witholding both chocolates.

However, while this pure strategy analysis is useful for gaining a basic understanding of the mechanisms of the game, it is much more realistic to model the game in extensive form, where players can change their strategies from turn to turn. In an extensive game, players are able to make a different choice at each round based on the information that they know (González-Dıaz, Garcıa-Jurado, & Fiestras-Janeiro, 2010). This extensive game framework better reflects the game we are describing, where players can choose to contribute a different amount of chocolate at each turn. Furthermore, in an extensive game we are also able to incorporate a varying temporal element (how many rounds until the end of the game), which allows us to explore the concept of discounting (e.g. players care less about the future reward the further away in time it is). Thus, when we build the model in the next section, we use an extensive game format.

Model Framework for the class game

In this section, we present a model framework proposed to study the game we played during the lectures in this semester.

Consider there are $\mathrm{N_g}$ number of groups with identical number of players N_s (in our case, $N_g = 4$, and $N_s = 9$). In the class experiment, the game was played at each class in 10 discrete rounds, therefore we consider the time setting to be discrete. Denote $x_k(t)$ as the number of points in the group account for group k at time t. We set $t = 1, 2, \ldots 10$, and $x_k(0) = 0$. At any observable and controllable time t during the course of the game, a typical player i within a typical group k ($i \in N_s, k \in N_g$) decides the number of points $a_i(t) \in \{0, 1, 2\}$ she will contribute to the group account. She makes her decision based on her knowledge about the group states after previous round[1] $x_k(t-1)$ and her own histories of dicisions $a_i(s)$ for all $s \leq t-1$, according to the following set of rules:

1). First, as data from the class game suggests that most participants considered their groups behavior before making the next decision, we assume player i makes her decision also depending on the inference of decisions made by other group members. Denote $a^{-}_{-i}(t)$ as the average number of points contributed by players other than i to her group account[2], we have $a^{-}_{-i}(t) = \frac{x_k(t) - \sum_{s=1}^{t} a_i(s)}{(N_s - 1) \cdot t}$. At time step t, player i decides her optimal choice $a_i(t)$ based on the observed $x_k(t-1)$ and the corresponding inferred $a^{-}_{-i}(t-1)$.

2). Second, player i aims to maximize her net payoff $p_i(t)$ at current time t. Her payoff comes from two parts, an immediate payoff $2 - a_i(t)$ and an expected future payoff amount with consideration of uncertainty. Denote the probability for group k to achieve the target of 90 points at terminal time T as $p_k(t) := \mathbb{P}(x_k(T) \geq 90)$. Player i calculates this probability by expecting $x_k(T) = x_k(t-1) + [a_i(t) + a^{-}_{-i}(t-1) \cdot (N_s - 1)] \cdot (T - t + 1)$, namely, the group account grows at a constant rate which comes from her as well as other group members' contribution to the group account. Therefore, $p_k(t) = 1$ whenever there is still chance for the group to reach the threshold (depending on its previous level

[1]Note here, the group states anounced at each time step t is actually the group state at time $t-1$.

[2]All other players are treated equally for play i since players are assumed to be homogeneous.

$x_k(t-1)$ and its projected growth rate). On the other hand, $p_k(t) = 0$ when, for example, its previous state $x_k(t-1)$ is low, and there isn't too much time left (t close to T).

3). Third, player i discount an amount of future reward according to the discounting function $\delta_i(t)$, it can take either hyperbolic or exponential format, i.e. $\delta_i(t) = e^{-r_i t}$ or $\delta_i(t) = \frac{1}{1+r_i t}$. In other words, 20 chocolates at the terminal time T equates $20 \cdot \delta_i(T-t)$ at the current time t.

Then player i makes her decision $a_i^*(t)$ at time t, where

$$a_i^*(t) = \begin{cases} 0, & \text{if } x_k(t-1) \geq 90. \\ \text{argmax}_{a_{i \in \{0,1,2\}}} u_i(t), & \text{otherwise.} \end{cases} \tag{1}$$

and

$$u_i(t) = \quad (2 - a_i) + 20 \cdot \delta_i(T-t) \cdot p_k(t). \tag{2}$$

The group account is updated according to the following rule:

$$x_k(t) = \quad x_k(t-1) + \sum_{j \in G_k} a_j(t). \tag{3}$$

Namely, we update the group account by summing over the real contributions of each player in that group.

Equation (1) says that when the group has already achieves the target, the future reward is certain and player i has no incentive to contribute to the group account any more. When the threshold has not been reached, however, player i determines her decision by maximizing her payoff function $u_i(t)$. Observing from (2), the payoff of player i comes from two parts, an immediate payoff and an expected amount of discounted future payoff. If her group has been quite cooperating, $p_k(t)$ equals to 1, thus boosting her confidence of getting the future rewards and motivating her to contribute more to the group account. On contrary, if there are majority of non-cooperators reflected in the low group account $x_k(t-1)$, then player i has less confidence of achieving the future reward, thus inducing her to maximize her immediate payoff. If this happens, then the initial non-cooperative behaviors of some group members will trigger a negative feedback loop that leads the group not to be

able to achieve the threshold eventually. Notice that (1) holds for every player i in a group, therefore, this model assumes identical behavior within a group. For example, it is reasonable for player i to choose $a_i(1) = 1$ initially, since she reason that all other players, by solving the identical optimization problem, will make the same decision. This strategy realizes a positive immediate reward as well as a positive future reward (because $p_k(1) = 1$). Yet, the model still supports a path-dependent solution, since player i is solving a different optimization problem as $x_k(t)$ varies from each group, and at each round.

Worth mentioning are the following assumptions made for this model framework. First, we assume homogeneity for all players across groups. Specifically, the payoff (or, utility) functions are considered to be the same for all players across groups. Moreover, we assume all other group members are essentially identical for player i, therefore she only needs to consider an average player's decision $a^{-}_{-i}(t)$. This assumption is reasonable since group members are assigned at random and in general, they do not know who are in their same groups. Moreover, the model considers scenarios only before reaching the threshold. As a result, more possible for the group to achieves the target increases one's weight on the future reward. The model is also supported by the empirical data in the following ways. First, both the data of game and exit survey indicated that the future payoff promised provided a large enough incentive for the majority of participants to contribute to the group account. Second, many participants considered the behavior of others before making their decisions, and this influence them in adjusting or changing their behaviors. therefore, the incorporation of $a^{-}_{-i}(t)$ is well-justified.

The model framework is designed to incorporate fewest possible numbers of parameters with a focus on examining the discounting factor. The model is expected to be supported by empirical evidence form the data of class games. Specifically, we hope to test if hyperbolic or exponential discounting is more applied for the class game, and to find out the appropriate range of parameter values for r_i.

For convenience, the parameters and variables used in this model are listed below:

N_g : number of groups

N_s : number of players in each group (size of a group)

$x_k(t)$: group account for group k at time t

$a_i(t)$: number of points player i decides to contribute to the group account at time t, $a_i(t) \in \{0, 1, 2\}$

$a^-_{-i}(t)$: average number of points contributed by players other than i to her group account, $a^-_{-i}(t) \in [0,$

$u_i(t)$: utility/payoff function for player i

$p_k(t)$: if it is possible for group k to meet the threshold

Finally, since we are essentially motivated to understand the collective decision making mechanisms for threshold public goods games in a much broader context, we discuss in table 1 some comparisons between this game and real world situations.

	Class experiment	**Real life cases**
Overall uncertainty	As it is typically the case for designed experiments, the conditions, rules and outcomes are explicitly specified in relatively simple and straightforward ways. There are no exogenous threat outside of the game system.	Yet, in real settings, it is often hard to quantify uncertainty. For example, it remains a difficult question to measure the cost of one Celsius increase in temperature, let alone being able to weigh the benefits and costs of different strategies to play in the game.
Threshold public goods	The simple numerical threshold in the class game is explicitly superimposed at the beginning of the game. It remains constant throughout the course of the game, and each group faces the same threshold.	Tipping points in nature emerges implicitly under complex uncertainties. In most real world systems, the following questions are often unknown. First, does a threshold even exist? If it exists, in what way? How will the threshold change along time and responding to human behavior? Finally, do different players face different thresholds?
Timescale and feedback	The class game last for an academic semester, which is relatively short compared with real life *games*. Moreover, the feedback (knowledge about the state) was quick, with only one-week lag.	Often, it takes longer time till realization of real collective system dynamics (time scale often occur in decades or longer). In addition, delays between feedback and response are often longer in natural and societal systems.
Interaction with others	In the class experiment, each player in the game is essentially regarded as homogeneous. In addition, there are no interactions either within or across groups. Finally, we observe interactions and outcomes at class level (single scale).	In reality, players (e.g. countries) have distinct characteristics that they usually cannot be regarded as homogeneous players. Moreover, players in real-world situations are connected with certain network structure. They have (at least) partial knowledge about their closely related partners, either as cooperators or competitors. Since players have incentives to make a better decision by having better knowledge about others in the game which is expected to yield better predictions of others' behaviors. Finally, interactions happen across scales at multi-levels (efforts are made in individual levels, corporation levels, and national levels).

Table 1: Comparison between the class game with real world situations.

Model Implementation

The model is implemented in Mathematica with exponential and hyperbolic discounting. Since the probabilities given in the previous section are always either 0 or 1, the threshold model is deterministic. With 3 possible choices and 9 players, the size of the state space is 19,683. However, since all players are identical, the number of unique initial states (that are not equivalent up to a permutation) is given by the Stirling number of the second kind, S(9,3)=3,025.

Random initial conditions

For convenience, instead of evaluating the trajectory of each of these unique initial states, we randomly draw 5,000 states from the set of states with uniform distribution. Then, the distribution of the final account value is plotted for 12 different values of r below.

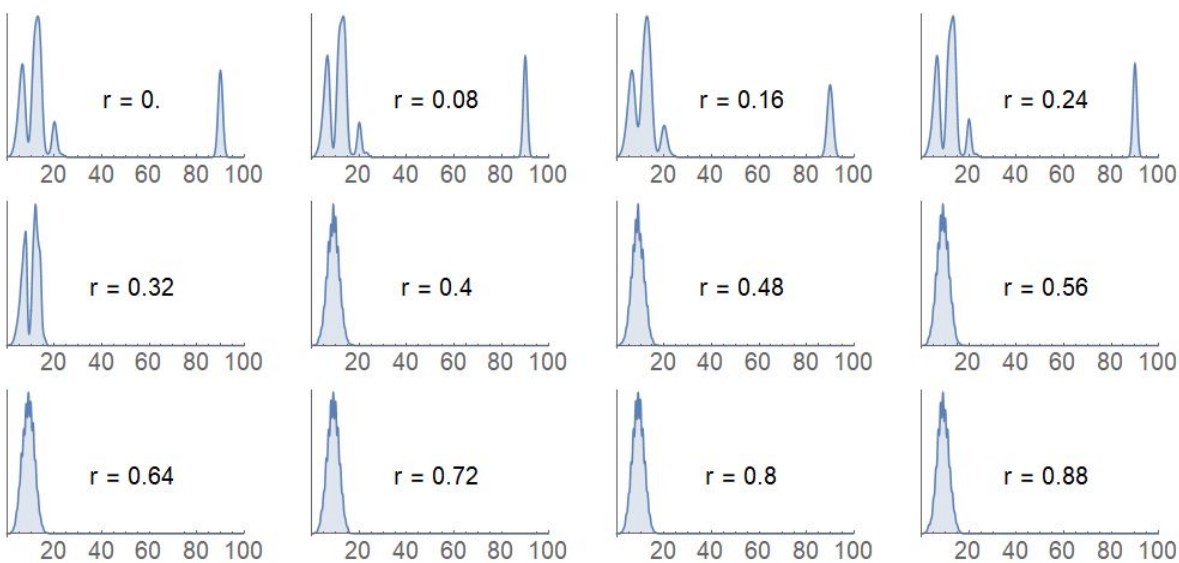

Figure 3: *Distribution of final accounts x(T) for many values of r with exponential discounting. The x-axis represents the final points in the account, the y-axis represents the frequency with which each x value occurs.*

In these plots, we can see that the peak on the right, namely the probability of x(T) = 90, vanishes suddenly between r = 0.24 and 0.32. To see how sharp this transition is, we can plot only the probability of x(T) = 90 as a function of r:

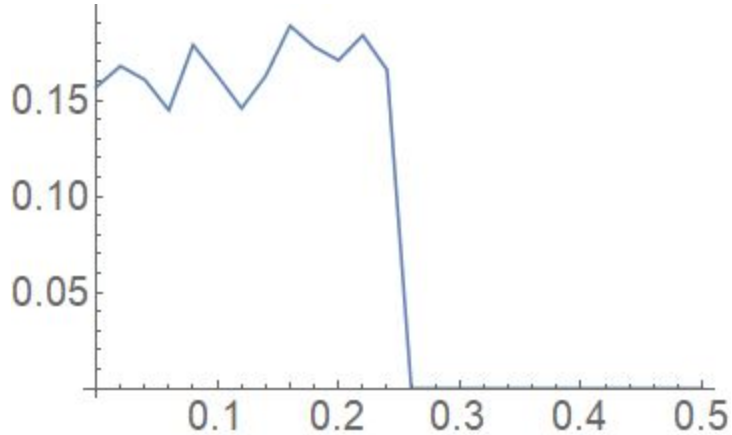

Figure 4: *The sharp transition from the region where reaching the goal (90) is possible to where it is impossible is around r = 0.25*

As discounting gets stronger, players choose not to contribute. This abrupt change is an artifact of the thresholding of p(t) in our model, which assumes 0 if the expected value of x(T) is smaller than 90 and assumes 1 otherwise.

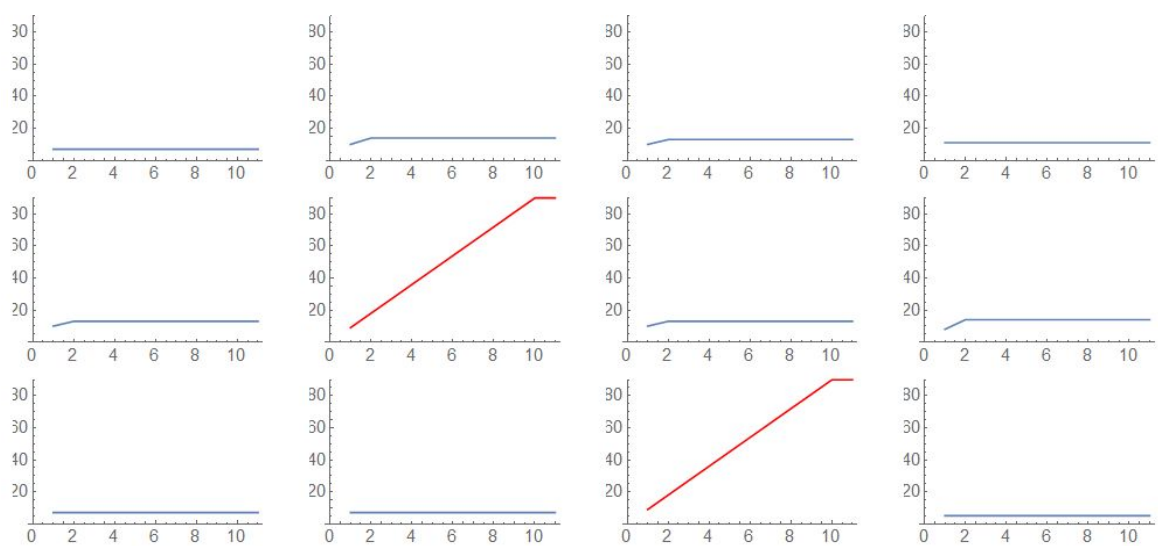

Figure 5: *12 random trajectories of x(t). The trials where x(T)=90 are colored red.*

Hyperbolic discounting

Everything else held constant, changing the form of the discounting function from exponential to hyperbolic removes this behavior.

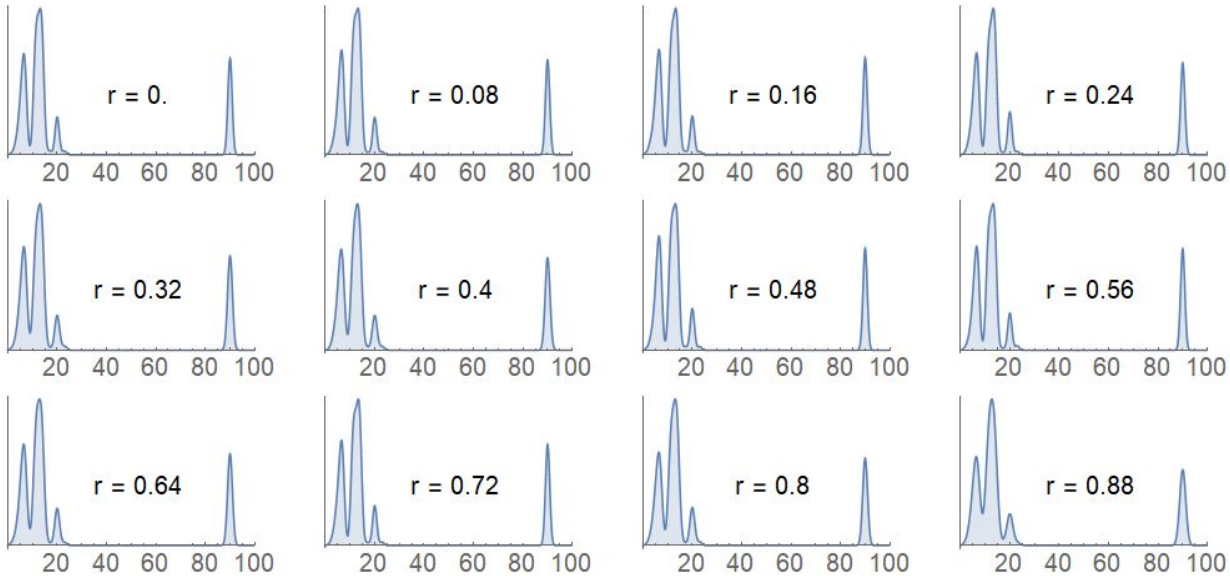

Figure 6: *Distribution of final accounts x(T) for many values of r with hyperbolic discounting. Again, the x-axis represents the final points in the account, the y-axis represents the frequency with which each x value occurs.*

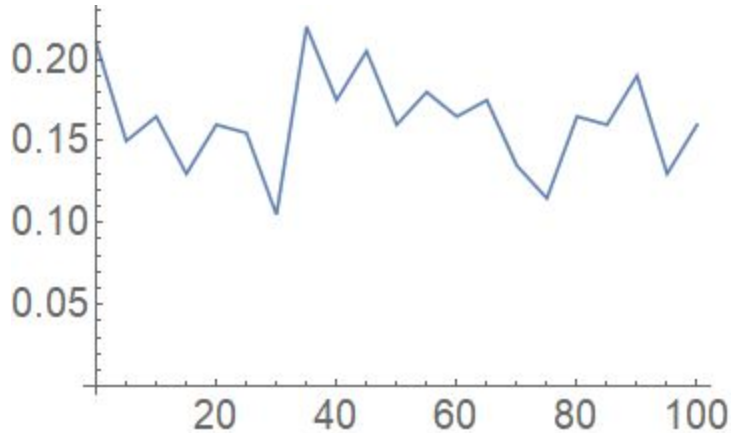

Figure 7: *The sharp transition has disappeared. Even at r = 100, which is much greater than the critical point r = 0.25 from the exponential discounting trials, there is no sign of transitioning.*

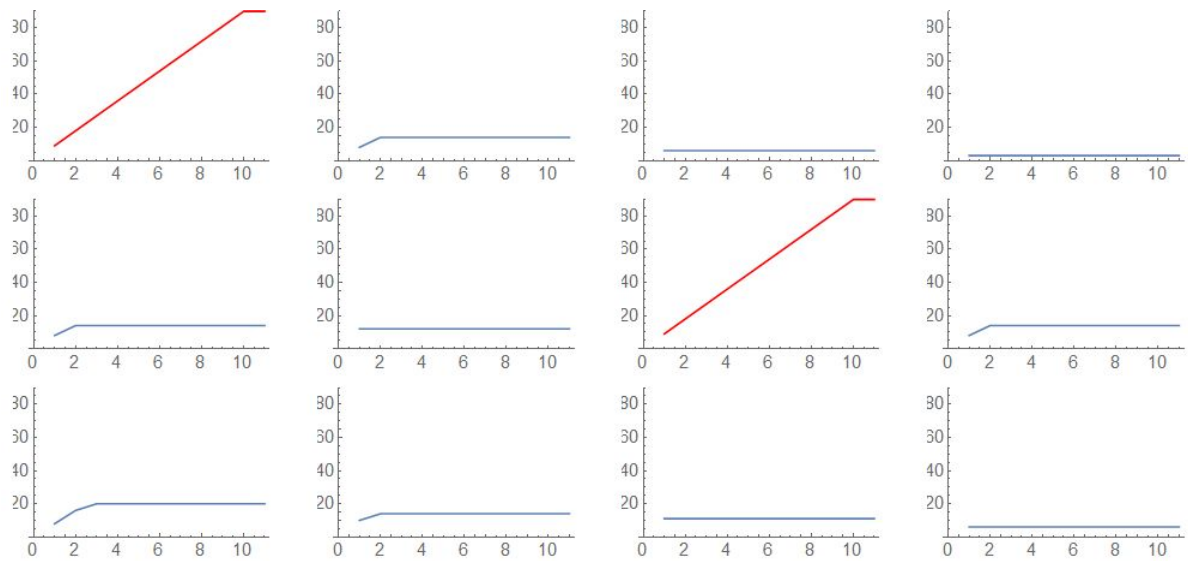

Figure 8: *12 random trajectories of x(t). The trials where x(T)=90 are colored red.*

Initializing with class data

Instead of exploring the whole state space, we can take the first recorded states from the class experiment. The initial actions of group A and group B were (2, 2, 0, 0, 0, 1, 1, 2, 0) and (2, 1, 1, 1, 2, 1, 1, 1, 1). Running the simulation with these initial conditions, we can plot x(T) against r:

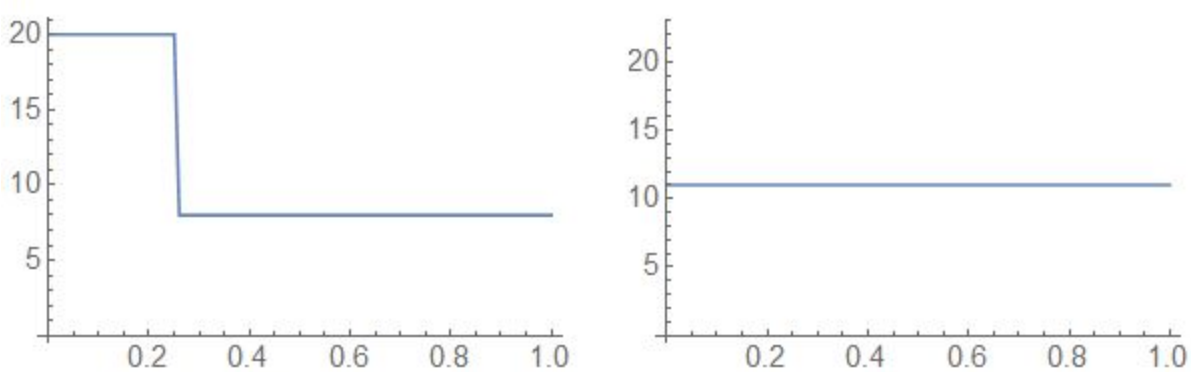

Figure 9: *Right: Group A. Left: Group B. These are results with exponential discounting. The x-axis represents discounting strength, the y-axis represents the final points in the account.*

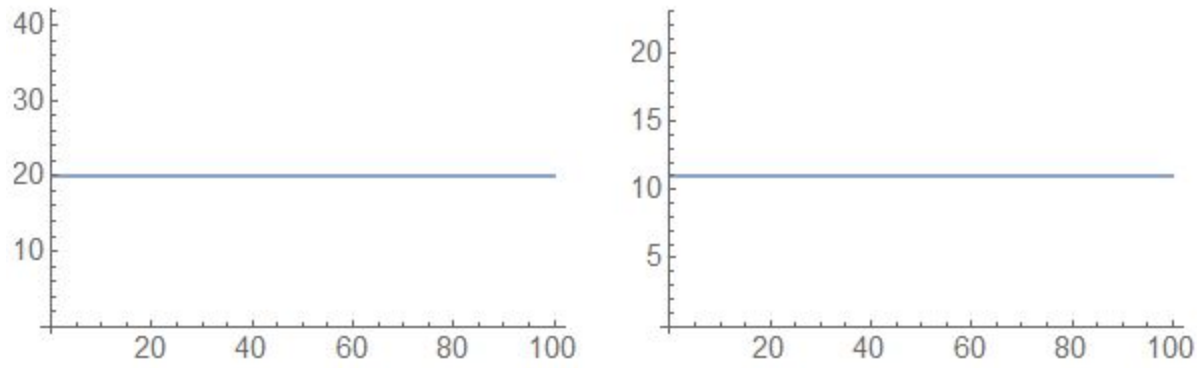

Figure 10: *Right: Group A. Left: Group B. These are results with hyperbolic discounting.The x-axis represents discounting strength, the y-axis represents the final points in the account.*

Here, we can observe the difference between exponential and hyperbolic discounting in group A simulations. In the hyperbolic case, despite raising r up to 100, the sharp drop from 20 to 8 of the exponential discounting is not observed. However, in contrast to the real values of x(T), neither model manages to reach the goal of 90 for any value of r. This reveals that our model is incomplete.

Survey Data

Analyzing the data from the class survey produced the following results. Most players (29/36) varied their strategy from turn to turn, so in terms of modelling actual human behavior, the pure strategies analysis is very limited. In 24/32 survey responses people said they considered the points in the group account when they made their decisions, supporting the mean-field framework (given that no players knew exactly how many points each other player put in, only the total contribution of the group per round. Only a minority of players did not like chocolate (1.25/35 response), suggesting that the chocolate was a good incentive/representative of a real public good that people desire. Most players said the rules were clear, and all players said the rules were clear or somewhat clear, supporting use of the game as a model. Finally, some students put “Fair play” as a main driver of their decisions, and students who checked “fair play” as a main driver of their decisions contributed on average about 2 points more than students who did not, suggesting influence of prosociality and not purely self-interest. (Data for this analysis comes from Kruger, Barfuss, Vasconcelos, 2020).

Discussion

Our motivating question was to understand the effects of time discounting in this threshold public goods game. But before we can understand the effects of time discounting on the game, we need to understand how the game works on a much more simple level when time discounting is not considered. From the pure strategies framework of the game (Figure 1 and 2), we see that this game has 2 equilibria in pure strategies where every player plays the same strategy and no players have incentive to change strategies (Figure 1). In this scenario, we have a defector equilibrium, where nobody contributes to the collective account, so no individual player wants to waste their chocolates to start contributing to the group account for no reward. The existence of this equilibrium is typical of public goods games, and it is this defector equilibrium that people trying to address public goods problems in real societal situations are concerned about avoiding (Rand et al., 2009). We also have a cooperative equilibrium, where every player contributes ½ of their chocolate to the group account, and since the group exactly reaches the threshold, no player wants to increase their contribution (which only decreases their payoff), and no player wants to decrease their payoff (since the additional payoff from defecting is less than the chocolates lost by the group no longer reaching the threshold). This is consistent with Bolle and with Wang et. al's point that Threshold public goods games generally have multiple equilibria, including cooperative possibilities (Bolle, 2014; Wang et al., 2009). However, it never increases any player's payoff to be fully altruistic and donate both chocolates to the group account.

When we allow every player to choose their own pure strategy independently, then plot the payoffs for a focal player choosing to contribute 0,1, or 2 chocolates (Figure 2), we get similar patterns on a more continuous graph. When the average contribution of the other players is over about 1.1, the strategy with the highest payoff is to defect and freeride off of others' contributions, so there is no equilibrium in this range. Similarly, when the average contribution of other players is below about 0.9, the best strategy is to defect and contribute 0 to the group account, since there is no chance of reaching the threshold. So in this range, everyone contributing 0 is again an equilibrium. When the average contribution ranges from 1 to 1.1, the best strategy is to contribute 1, which corresponds to the cooperative equilibrium in the paragraph above. From 0.9 to 1, it is equally beneficial to defect or to contribute 2, suggesting that with a very small amount of defection it is still possible to have cooperative behavior, but that if even one player defects, there is no incentive for other players to pick up the slack with their own donations. Now with this basic understanding of the game, we can see what happens when we bring in discounting.

The extensive game model expands on this basic game theoretical understanding of the game, while incorporating the time-delayed reward, and allowing a more realistic framework for player behavior where players can change their strategy at every turn. This allowed us to address our goal for this study, which was to understand the role and effects of time discounting on cooperation in this threshold public goods game.

We found that as the strength of discounting increases, for exponential discounting, the probability that a team with random starting conditions will be able to reach the threshold overall decreases: at low levels of discounting, there is some chance that the team will behave according to the cooperative equilibrium, but at higher levels of discounting, the team is never able to cooperate to reach the threshold (Figure 3). Interestingly, in the exponential condition, as discounting increases, this change does not occur gradually, but instead once discounting strength crosses a threshold at about r=0.25, the chance of reaching the threshold drops from a previously fairly stable level immediately to 0.

When we considered hyperbolic discounting, the pattern was not so distinct. Overall, the chance that a group will cooperate appears to be decreasing, the chance of reaching the threshold was lower at higher discount levels than at very low levels (Figure 7). Though there appears to be some variation at extremely high r values, at more socially realistic levels of discounting (Figure 6,7) there appears to be a consistent decrease in the probability of cooperation from lower to higher levels of discounting. Notably, under hyperbolic discounting there is always some probability that the group will manage to cooperate, unlike the exponential case there is no threshold above which no cooperation occurs. This is likely because under hyperbolic discounting, after a rapid decrease in valuation, discounting for long term delays actually increases less steeply. So in the hyperbolic case, the long term delayed reward seems to continually retain enough value that cooperation is at least theoretically possible, whereas in the exponential case, the discount for long-term rewards does not become less steep, so after a certain point, the long term reward basically has negligible value, and players will focus only on the short term rewards. This general decrease in group cooperation with higher discounting is consistent with past studies that have found increasing defection in two-player prisoner's dilemma games with higher temporal discounting (Harris & Madden, 2002).

We compared the model data to the class data (Figures 9,10), and for both types of discounting the real groups outperformed the model: Group A had 64 points in the class account and Group B had 98 when the game ended. However, in the model, neither group achieves the final goal. This suggests that,

by choosing to focus exclusively on the behavior of temporal discounting, our model likely ended up missing other factors that influence the behavior of real groups. For example, we did not consider the influence of prosociality. Prosociality, or the tendency of individuals to take others into account, is well documented in humans (Jin, Pei, & Ma, 2017) and could account for people donating more points than would be expected if they were only interested in personal gain. Notably, in the survey data we see that there are players who put "Fair Play" as their main motivator, and these students contributed more points to the group account than students who put something else as their main motivator. This demonstrates that students in our class definitely had a sense of prosociality (probably even the students for whom it was not the main motivator) and also that this prosociality appreciably altered their behavior. The incongruity between the model and the experimental data might also have been exacerbated by our small sample size--we only had a record of 2 groups' behavior, while we would want data from many more groups to have a more robust comparison.

In this project, our model framework was designed to incorporate the fewest possible parameters, focusing on examining the specific discounting factor. Under the premise of identifying which discounting function was more prominent among players, we assume that all players within the game could be generalized, played rationally and according to their optimal preference. With respect to the utility and payoffs, we assumed that there was absolute certainty of payoffs, that is chocolate/fruit was provided immediately when a player designated for him/herself, and that all players were aware that if they met the 90 threshold, they would receive 90 additional chocolates/fruit as a group with certainty. It was also assumed that all players derived positive utility from the rewards of the game (chocolate/fruit), and that all rewards for the game were meant for the individual themself and not shared after. Survey data supports these assumptions: the vast majority of players liked chocolate, found the rules of the game clear, and considered the points in the group account/others' behavior before making decisions. With respect to interaction with other players in their group, or in other groups we assumed that players had no interactions that could have caused complications in decision making (no collusion), contrary to the practical applications of player interactions in the following paragraphs. In the context of the game, this makes sense since players did not know or communicate with their group mates. Lastly, with respect to players, we assumed there was sufficient homogeneity within our test subjects that we could extrapolate patterns or generalizations in their behavior. All players in the game were of equal importance and had equal opportunity and fairness in their decisions.

In summary, higher discount rates generally hinder cooperation as individuals who weigh future

payoffs or utilities would be more inclined to take now for themselves. Additionally, in societal problems, individuals are often incentivized to compare their share or wealth relative to others. This was not necessarily accounted for in the context of this game where knowledge of other players' contributions and taking were mostly private. Likewise, for public goods games in the real world, discount rates and value of payoffs are often subjective. Since the discount rates are subjective, players might have varying discount rates, which would affect how they view future payoffs, as higher discount rates would equate to a lower present value for future payoffs, and more incentive to take now, than contribute to the societal fund. An additional constraint to consider is the fact that players may have certain liability needs, or require certain payments be met at various points in time, which would also influence their decision to prolong utility. Additional research that future researchers could conduct would be to uncover differences in initial and final behavior in achieving the group target, as this was evident in the data from the class experiment. Similarly, we could investigate players' strategy change in response to probabilistic conditions of achieving the goal, looking at a more quantifiable method of understanding player behavior on an inter-week basis. Lastly, we could measure the prosociality and spite among various players in the game, motivated by patterns we found in consistent group contribution.

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